



IDBI INNOVATE 2026

Build. Integrate. Transform.

Knowledge Partner



Powered by



Technology Partner



Team Details

- a. **Team name:** Lambdac Computing
- b. **Team leader name:** Reshma Mohan
- c. **Problem Statement:** Problem Statement 5 - Open Innovation

Brief about the idea

CreditPulse - Fraud Intelligence: explainable mule-account detection with citation-gated case files

- The pick: Problem Statement 5 invites out-of-the-box submissions. We chose the fraud problem every Indian bank is being directed to solve: mule accounts and fraud rings on the payment rails (MHA directive for FIs to integrate mule detection by December 2026; RBIH MuleHunter.AI momentum; RBI FREE-AI explainability principles).
- The problem: a mule ring drains an account in minutes. Freeze too slowly and the money is gone; freeze too eagerly and an honest customer is locked out.
- The idea: score every account with behaviour-pattern detectors plus an independent anomaly leg, expand a suspicious account into its ring across the transaction graph, then assemble a five-stage, citation-gated case file where every claim carries the transaction IDs behind it. An uncited claim raises an error rather than rendering.
- The focus: keeping false positives down on the exact accounts that get wrongly frozen in production - high-velocity but honest accounts. 0 of 10 hard negatives flagged, each cleared with reasons.

Note: this entry is one track of CreditPulse, one codebase and one deployment answering three problem statements (PS3 Financial Health Score, PS4 Early Warning / Default Prediction, PS5 Fraud Intelligence).

Opportunities

How different is it from any of the other existing ideas?

- Anomaly-score entries either miss coordinated rings (each account looks individually fine) or freeze honest high-velocity customers. CreditPulse catches the structure: ring-level recovery over the transfer graph - 6 of 6 rings on the holdout.
- The citation gate is the key idea: a case cannot be built without its evidence, so investigators open a pre-sourced dossier, not a bare score.

How will it be able to solve the problem?

- Runs on data the bank already holds (accounts and transfers): no new data source, no graph database, no new dependency - ring expansion is a bounded pure-Python graph walk, fast and auditable.
- Human-in-the-loop by design: the system recommends (freeze and STR draft / enhanced monitoring / clear with note); the analyst approves or overrides; every decision lands in an audit trail.

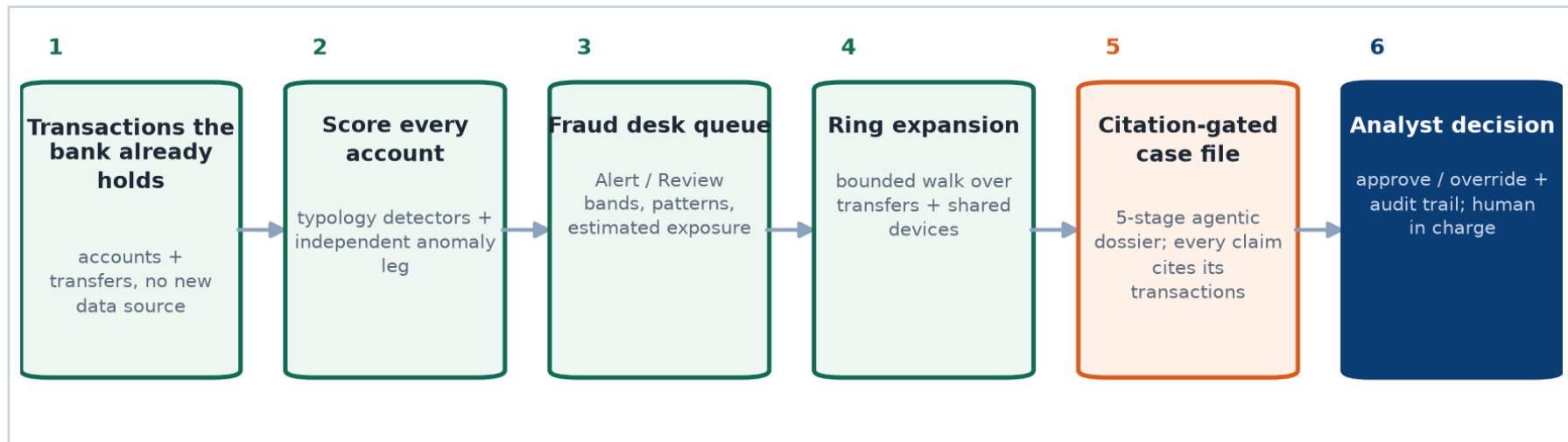
USP of the proposed solution

- 6/6 fraud rings recovered with 0/10 hard-negative false positives, and a case file the bank can defend line by line - to a customer, a regulator, or a court.

List of features offered by the solution

- Fraud Desk: a scored account queue with Alert / Review / Clear bands, behaviour-pattern chips and estimated exposure
- Eight behaviour-pattern detectors: fan-in / fan-out, rapid pass-through, dormancy burst, new-account velocity, KYC-income mismatch, round-amount structuring, odd hours, shared device
- An independent anomaly leg (Isolation Forest) blended with the pattern score - two legs, disclosed separately on screen
- Ring expansion: a bounded graph walk over repeated transfers and shared devices that recovers the ring structure, not just accounts
- Five-stage agentic investigation (triage, evidence, network, adjudication, case-file compiler), animated live in the product
- Citation gate: every ground of suspicion opens the exact transactions that triggered it; an uncited claim raises
- Hard-negative discipline: honest high-velocity accounts are explicitly cleared, with reasons on screen
- Analyst decision gate: approve / override with a deterministic audit trail; STR-draft recommendation where warranted
- Ground-truth labels are evaluation-only and never read at score time - enforced structurally and by tests

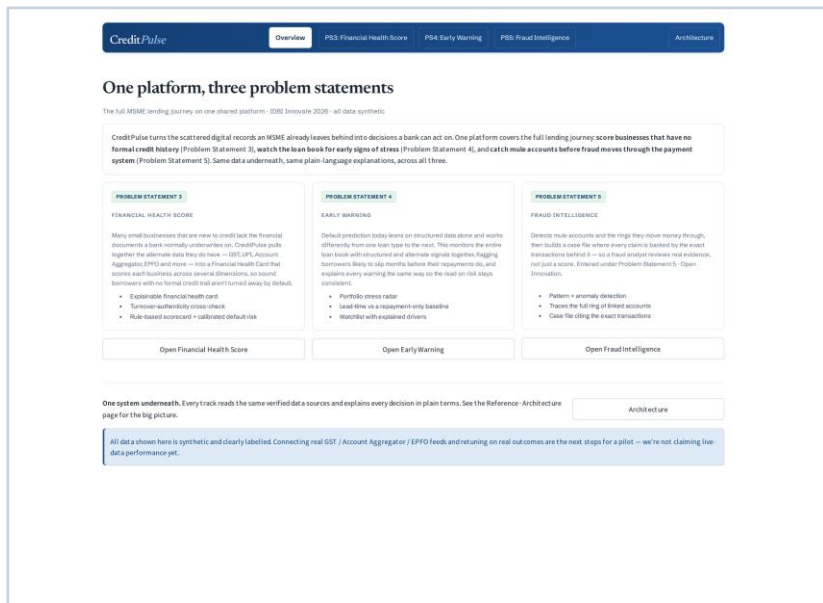
Process flow diagram or Use-case diagram



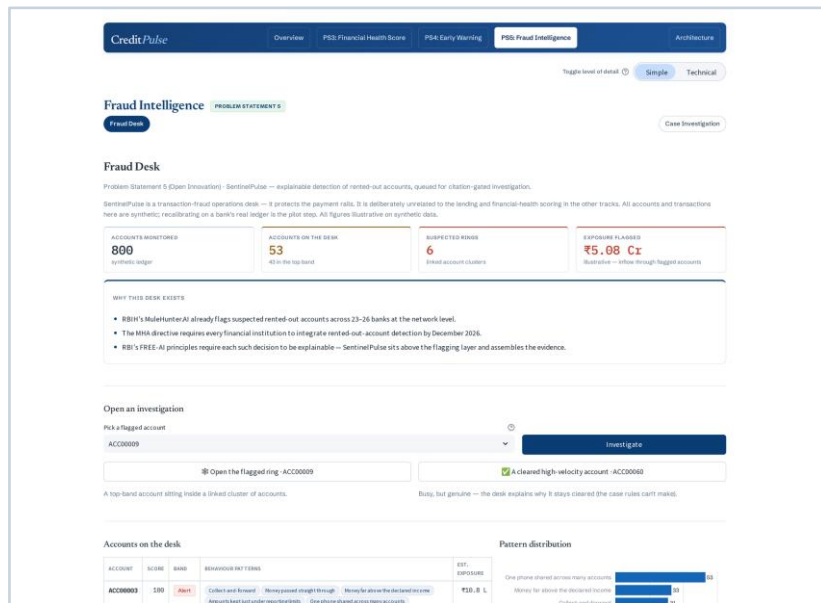
From raw transfers to a defensible decision: detection, ring recovery, a citation-gated dossier, and a human call - recorded, every time.

Wireframes/Mock diagrams of the proposed solution (optional)

The prototype is implemented and deployed; actual product screens below stand in for wireframes.

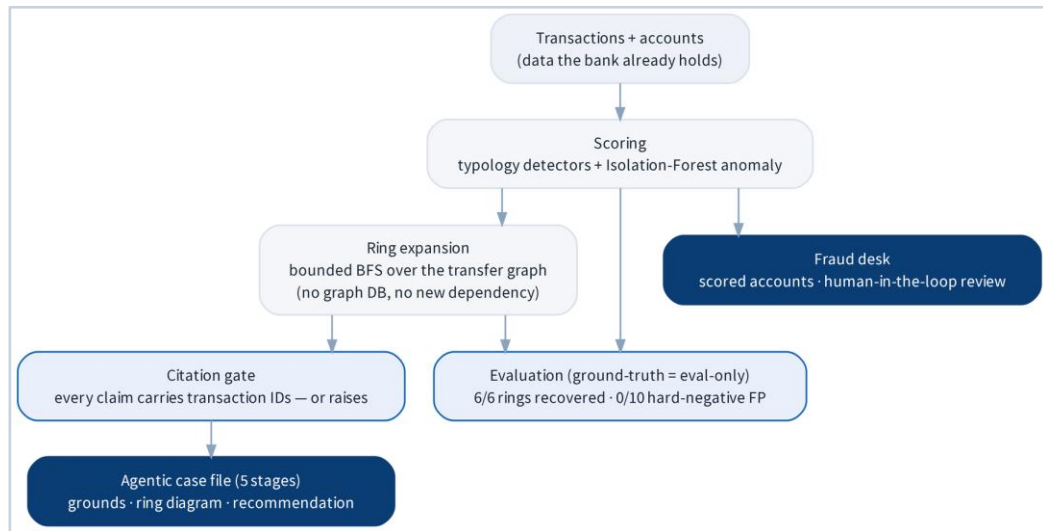


Platform landing page: one card per problem statement



Fraud Desk: the scored queue with bands, patterns and exposure

Architecture diagram of the proposed solution



Rendered live inside the product (Architecture page, Problem Statement 5 view).

Self-contained track

- Data generation, detectors, engine, orchestrator and pages live in one folder; imports only the shared platform core

No heavyweight machinery

- No graph database, no GNN, no runtime LLM: bounded pure-Python ring expansion and deterministic agent stages keep every step auditable

Citation gate

- The case-file compiler refuses to show a claim without its transaction IDs - the evidence check is built in, not a manual step afterwards

Deployment

- Same container and deploy as the other tracks; deep link /track05

Technologies to be used in the solution

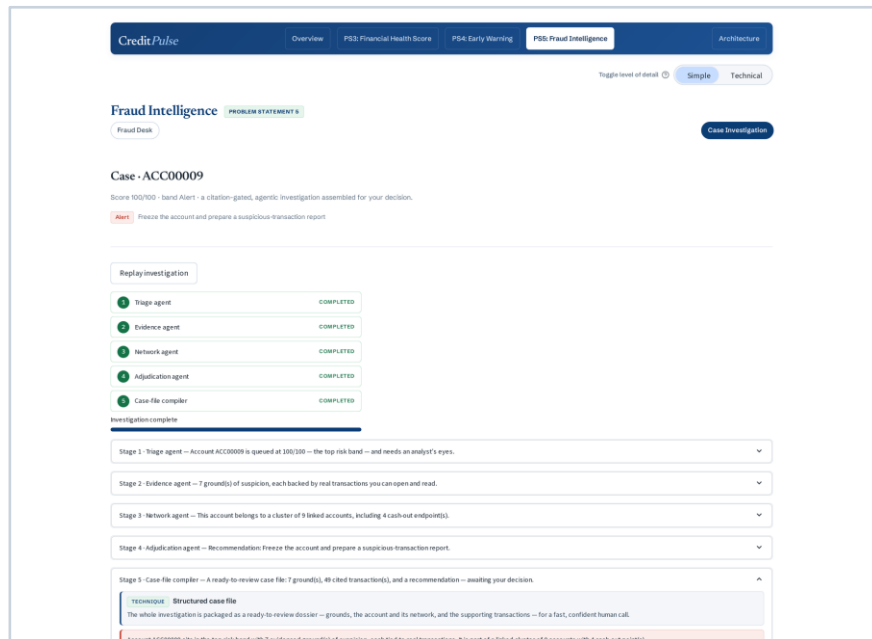
- Python 3.12; pandas + NumPy (data layer), scikit-learn (scorecard, calibration, clustering), LightGBM (monotonicity-constrained gradient boosting), SHAP (explainability)
- Streamlit multi-page product (st.navigation router); Plotly charts; Graphviz architecture diagrams
- Deterministic synthetic-data generators; transaction typologies modelled on documented Indian mule-account patterns; fixed seeds, identical output on every run
- Docker single-container deploy (Cloud Run today, portable to the bank's AWS / ACC sandbox at stage 2); GitHub Actions CI regenerates data, runs the full test suite and eval harness, and builds the image on every push
- pytest: 296 tests covering determinism, monotonicity, leakage guards, UI smoke sweeps and track isolation
- No runtime LLM and no black-box service calls: every line of text on screen is assembled from the underlying computation, so the product can be audited end to end
- Track-specific: a typology-driven synthetic transaction generator (6 planted rings, hard-negative gig workers, 175k+ transfers) and an agentic case orchestrator with a structural citation gate

Estimated implementation cost (optional)

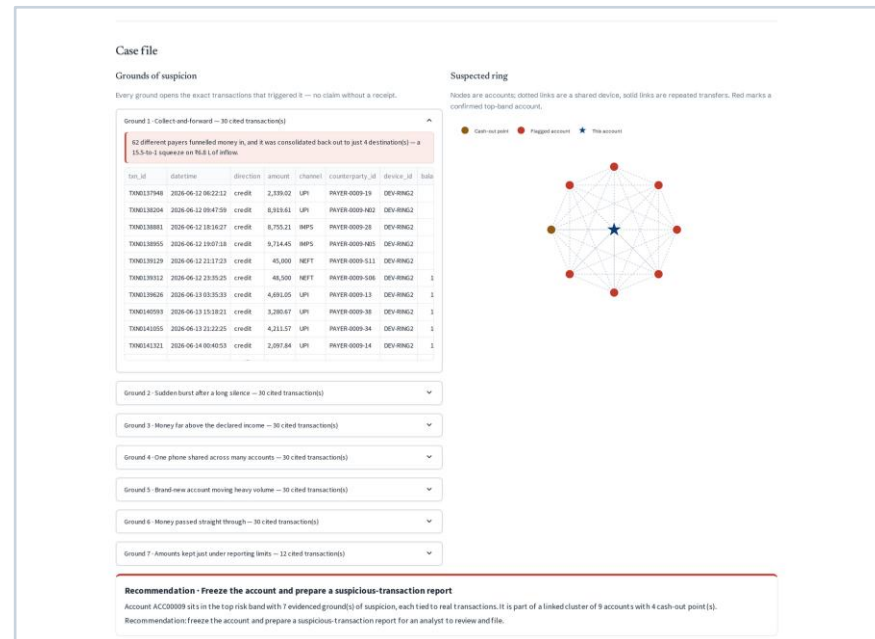
Indicative figures, stage-1 prototype scope.

- Open-source stack end to end: no licence fees, no per-score charges
- One small container (2 vCPU / 2 GB, scale-to-zero) serves all three problem statements; indicative cloud cost well under Rs 5,000 per month at demo traffic
- Pilot-phase cost is engineering, not infrastructure: AA / ULI adapter integration, DPDP consent flows, security review, and model recalibration on bank data
- Scoring is stateless and batch-friendly: portfolio-scale scoring adds compute linearly, with no specialised hardware

Snapshots of the prototype



Case investigation: the five-stage agentic run for a flagged account, narrated live



The case file: grounds of suspicion with cited transactions, and the suspected ring recovered from the transfer graph

Prototype Performance report/Benchmarking

Detection scorecard - synthetic holdout (800 accounts, 32 mules, 6 rings)

- Ring recall: 6/6 rings recovered (a ring counts as caught only if at least 60% of its members are flagged)
- Hard-negative false positives: 0/10 - busy-but-honest accounts stay clear, with reasons
- Alert level: recall 1.00, precision 0.74; ring-level precision 1.00 - every alert is genuinely ring-associated infrastructure (recruiters, cash-out points)
- Per-typology capture: six of eight patterns at 1.00 (fan-in / fan-out 0.81, rapid pass-through 0.44 - those accounts are still caught by the blended score's other legs)
- Ground truth is evaluation-only: the engine and the pages never read the labels at score time - enforced structurally

Synthetic transaction data; typologies modelled on documented Indian mule-account patterns. Live-rail recalibration and alert-volume tuning is the pilot step.

Additional Details/Future Development (if any)

- Pilot on live rails: streaming scoring, alert-volume tuning against investigator capacity, and a feedback loop from analyst decisions into thresholds
- STR workflow: auto-draft suspicious-transaction reports from the case file - the evidence is already assembled and cited
- Integration hooks for consortium signals (RBIH MuleHunter.AI style) and inter-bank ring visibility
- Extend the typology library - new patterns are additive detectors, not retraining events; DPDP-aware retention policy for case data

Provide links to your:

- GitHub Public Repository
- Demo Video Link (3 Minutes)
- Final Product Link

GitHub Public Repository:

<https://github.com/lambdacc/idbi>

Demo Video Link (3 Minutes):

<https://youtu.be/iMOVbhE0BL0>

Final Product Link:

<https://creditpulse-66armtf3tq-el.a.run.app/>

Note: On the initial load of the demo deployment, the application may take a few seconds to appear. This is due to an on-demand startup mechanism used to minimize cloud hosting costs during the evaluation period. Once the application has loaded, please select the appropriate Problem Statement tab.

Data honesty and guardrails

- All data in the prototype is synthetic: generated, clearly labelled on screen, and modelled on documented Indian mule-account patterns. Ground-truth labels exist for evaluation only and are never read at score time.
- Built to resist gaming: monotonicity constraints (a worsening risk factor can never improve a score), leakage-resistant holdouts, and determinism tests, all enforced in the 296-test suite
- Every user-visible number is computed from the in-repo eval harnesses at build time, or is labelled illustrative
- Human-in-the-loop: the system recommends, the officer or analyst decides, and every decision leaves an audit trail
- What we do not claim: real-world accuracy. That is measured at pilot, on bank data, after recalibration.



IDBI 2026 INNOVATE

Build. Integrate. Transform.

Knowledge Partner



Powered by



Technology Partner



Thank you!